

GIC Labs:
Improving
meeting
efficiency via
timely

transcription
and translation

with Speech-

to-Text and

Translation API

Tell us your challenge. We're here to help.

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ABOUT GIC LABS



About GIC Labs

GIC Labs increases
knowledge retention
turns around foreign
language documents
quickly with Speech
for transcription and

GIC Labs is GIC's in-house innovation unit which accelerates the use of innovative technologies and incubates new ideas, in collaboration with business units. It undertakes research on technology trends and prototype solutions that can be implemented enterprise-wide. GIC is a leading global investment firm established in 1981 to manage Singapore's foreign reserves. As a disciplined long-term value investor, GIC is uniquely positioned for investments across a wide range of asset classes, including equities, fixed income, private equity, real estate, and

Translation API for translation.

infrastructure. Headquartered in Singapore, GIC has investments in over 40 countries and employs over 1,700 people across 10 offices in key financial cities worldwide.

Google Cloud results

- Reduces hours of manual transcription of meeting notes to just minutes with Speech-to-Text
- Automates manual tasks such as note-taking so employees can focus on higher value-added activities such as data analysis and delivering actionable insights
- Improves turnaround

Up to 10x faster transcription speed for more insights

Industries: Financial Services & Insurance

Location: Singapore

Speech to Text

Google Cloud (<https://cloud.google.com/>) and

Translation API

Cloud Vision API (<https://cloud.google.com/vision/>)

Cloud Speech to Text

(<https://cloud.google.com/speech/>)

Cloud Translation API

(<https://cloud.google.com/translate/>)

Contact Center AI

(<https://cloud.google.com/solutions/contact-center/>)

Contact us (v/contact)

time in
understanding
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Employees in global organizations spend 35% of their 40-hour workweek on meetings

(<https://www.bcg.com/publications/2020/maximizing-productivity-meetings-and-emails>)

. But how much of the information do they actually retain? At GIC (<https://www.gic.com.sg/>), a global investment firm for Singapore's foreign reserves, meetings happen every day. From management discussions on portfolio performance to training seminars for new employees, meeting participants often take notes to remember key discussion points and inform other team members about action items.

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AI are pre-trained on huge datasets for accuracy and efficiency. These solutions enable us to meet our objectives faster, since we do not have to start from scratch."

*—Brian Lim, Senior Vice
President, Head of GIC
Labs, GIC*

While some meetings may be confidential and cannot be recorded, many other meetings or audio sources are public in nature. To enable participants to quickly reference transcripts from public seminars or external audio feeds such as podcasts, Brian Lim, Senior Vice President and Head of GIC Labs, an in-house innovation unit within GIC, partnered with Google Cloud

(<https://cloud.google.com/>) to come up with a novel solution. By developing a web interface using Google Cloud artificial intelligence (AI) products and services like Vision AI (/vision), employees can use Speech-to-Text (/speech-to-text) API, part of Contact Center AI (/solutions/contact-center) solution, to

quickly transcribe non-confidential audio recordings into text.

“Google Cloud AI solutions such as Speech-to-Text and Vision AI are pre-trained on huge datasets for accuracy and efficiency,” says Brian. “These solutions enable us to meet our objectives faster, since we do not have to start from scratch. The speech recognition model also continues to improve over time with minimal additional data or tuning from us.”

“Speech-to-Text is able to meet most of our needs, across a broad range of scenarios. For example, instead of taking hours for a manual transcription, a rough transcript is ready much more quickly, and this has helped to

increase GIC
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efficiency.”

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Enlisting AI to improve translation efficiency

The GIC Labs team considered various factors when choosing a suitable transcription service to meet the needs of specific teams within GIC. Apart from turnaround time and cost, the team also looked at the word error rate to measure the transcription accuracy and speaker diarization to recognize multiple speakers in the same audio clip.

GIC Labs tested a range of transcription services with the same set of audio clips reflecting real-life scenarios, ranging from a presentation in an auditorium to a discussion between multiple speakers in a meeting room. Overall, it found Speech-to-Text to be fairly accurate, with a comparatively shorter turnaround time compared to other solutions.

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Customizing to user requirements

To transcribe an audio recording, employees upload the audio file to an internal web interface with customized features to meet user requirements. For example, once the transcription is completed, the end-user receives an email alert to review and download the text file from the web interface.

GIC Labs leverages built-in Speech-to-Text functionality for speech recognition to improve the transcription quality by filtering stop words such as, "I mean" and "um" in text results. The team also uses the speech adaptation feature to boost accuracy by helping the machine learning model learn frequently occurring words such as "GIC."

Taking automated transcription further, GIC Labs introduced color coding to indicate the confidence level of accuracy for individual words. Utilizing the built-in word confidence feature in Speech-to-Text, words with low accuracy are highlighted so users can quickly review highlighted areas to ensure these words are transcribed accurately. When the user clicks on a highlighted area, they can listen to the corresponding audio to edit the text instead of listening to the entire transcript.

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Working smarter with man and machine collaboration

Brian says that AI helps to augment human capabilities to be more productive. Besides Speech to Text, GIC Labs also utilizes other solutions in the

Google Cloud AI suite. GIC Labs uses Vision AI coupled with Translation API (/translate) to automatically extract and translate text from image-heavy documents in various languages to help employees prioritize their time on key documents.

“By combining Vision AI with Translation API, we can quickly understand foreign language documents and determine whether there is useful information to be extracted for further analysis,” says Brian. End users have also shared that the AI platform is a time-saver, since they can quickly decide whether a document requires a closer read.

“Google Cloud AI solutions such as Speech-to-Text minimize routine work such as note-taking and allows our team to focus on higher value-added activities such as analyzing trends and delivering actionable insights,” he adds.

